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Enhancement of the lithium-ion conductivity of $LiBH_4$ by hydration

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ABSTRACT

Lithium borohydride ($LiBH_4$) is a candidate material for a solid electrolyte of all-solid-state lithium secondary batteries. In this study, the hydration behavior and influences of hydration on the ionic conductivity of $LiBH_4$ were investigated. $LiBH_4$ was found to become hydrated without decomposing; structural water can dehydrate at 55 °C and then rehydrate at around 35 °C. Hydrated $LiBH_4$ has a high lithium-ion conductivity of 4.89×10^{-4} S · cm^{-1} at 45 °C. The temperature dependence of conductivity and motional narrowing of the ^{7}Li NMR spectra at around dehydration temperature suggest that the enhanced lithium-ion conductivity is attributed to the motion of structural water molecules.

1. Introduction

Lithium secondary batteries are widely used for mobile electric products because of their high energy density, lack of memory effect and low self-discharge rate [1–3]. However, the flammability and volatility of organic liquid electrolytes used in batteries have the potential to cause problems. In addition, organic liquid electrolytes set a limit to improve battery performances when higher capacity and larger scale batteries are required for the practical use of renewable energy.

Recently, all-solid-state lithium secondary batteries have been the subject of much attention as alternatives to batteries using organic liquid electrolytes [2]. Improvements in the safety and shape flexibility of all-solid-state lithium secondary batteries are likely with ongoing research. It has been shown that when only the lithium ions move in the solid electrolyte, side reactions are likely restrained and both the safety and battery life are improved.

Among the various solid electrolytes studied [4–11], lithium borohydride ($LiBH_4$) is attracting much attention [12,13]. $LiBH_4$ has an orthorhombic phase (space group: *Pmna*) at room temperature and it is an insulator. However, the hexagonal phase ($P6_3mc$) that results from a phase transition at 113 °C is characterized by high lithium-ion conductivity (10^{-3} S · cm^{-1}). While the high transition temperature is a definite obstacle to using $LiBH_4$ in batteries, it is possible to decrease the transition temperature of $LiBH_4$ by lithium halide doping, with the added benefit of increasing the ionic conductivity [14]. Especially, LiI-doped $LiBH_4$ has a high lithium-ion conductive phase at R.T. An all-solid-state lithium secondary battery using $LiBH_4$ and $LiCoO_2$ has been developed [15]. The cell kept 97% of the initial discharge capacity even after 30 charge–discharge cycles.

Another known characteristic of $LiBH_4$ is that it easily reacts with water. Hydration reactions of $LiBH_4$ along with hydrogen gas emission have already been reported as a hydrogen storage medium [16–19]. These reports focus on the hydrogen emission and decomposition behaviors. As yet, however, the relationship between hydration and the electrochemical properties has not been clarified from the point of view of an electrolyte. Knowledge of how the reaction with water affects the performance of $LiBH_4$ as an electrolyte is important if it is to be successfully applied to lithium secondary batteries. In addition, the absorbent character of some dopants may likewise affect the electrical properties of $LiBH_4$. In this paper, we discuss the effects of hydration on the lithium-ion conductivity of $LiBH_4$.

2. Experimental

A total of 50 mg of $LiBH_4$ powder (Aldrich, purity >95%) was divided into 3 and placed in cases (ϕ3 mm) made by a stainless steel wire mesh (500 mesh, aperture of 0.026 mm). These cases were enclosed in a glass tube and heat-treated in a vacuum at 100 °C for 5 h. Water vapor was then gradually introduced from $0.1P_0$ to $0.9P_0$ (P_0: saturation vapor pressure) at 25 °C by Sieverts apparatus (BEL Japan, Inc.; BELSORP 18) without exposure to air.

The crystal structures were determined by analysis using an X-ray diffraction meter (Bruker AXS; D8 advance) with Cu-Kα at 40 kV and 40 mA. The lattice constants of the samples were refined by WPPD (Whole Powder Pattern Decomposition) based on the Pawley method, using TOPAS 4 software. A differential scanning calorimetry (DSC) analysis was carried out using a DSC instrument (TA Instruments Japan Inc.; Q2000) in order to evaluate the stability of the hydrated samples.

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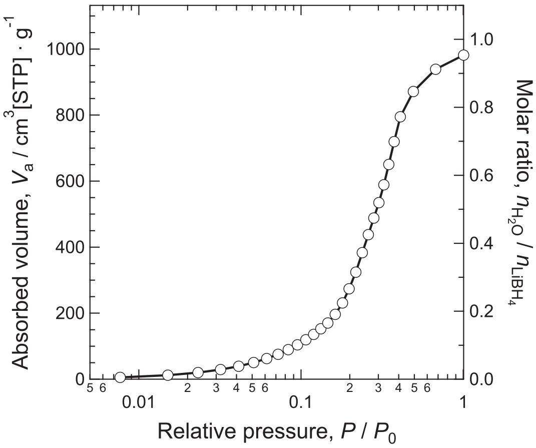

Fig. 1. $P - C$ isotherm of $LiBH_4$ at 25 °C for water vapor adsorption. The left axis shows a gas volume, V_a in standard temperature and pressure, the right axis shows a molar ratio of H_2O and $LiBH_4$, n_{H_2O}/n_{LiBH_4}.

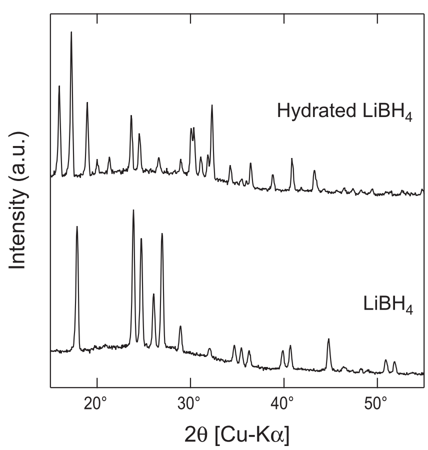

Fig. 2. X-ray diffraction patterns of hydrated $LiBH_4$ and dried $LiBH_4$. Hydrated $LiBH_4$ has a monoclinic structure ($P2_1/c$) and $LiBH_4$ has an orthorhombic structure ($Pnma$).

Samples were enclosed in an aluminum hermetic pan under a dry argon atmosphere. The temperature was changed at a rate of 5°C · min^{-1}.

AC impedance measurements using an impedance/gain-phase analyzer (Solartron; Model 1260A) and DC potentiostatic measurement using multichannel potentiostat/galvanostat instruments (Solartron; Model 1470E) were carried out under an Ar atmosphere in hermetically sealed vessels. The sample was pressed into a pellet (ϕ8 mm, ~1 mm-thick) and symmetrical cells were prepared using lithium or molybdenum foils as electrodes. ^{7}Li nuclear magnetic resonance (NMR) was conducted with ECA-300 (JEOL) at 7 T on 116.8 MHz. The chemical shifts were referenced to LiCl. The sample was heated in an argon flow.

3. Results and discussions

3.1. Hydration behavior of $LiBH_4$

Hydration behavior was investigated by Sieverts apparatus. Fig. 1 shows the $P - C$ isotherm of $LiBH_4$ at 25 °C for water vapor adsorption. The volume of absorbed water vapor increased with increasing vapor pressure until the molar ratio of H_2O and $LiBH_4$, n_{H_2O}/n_{LiBH_4} became saturated at around unity. Considering the amount of water vapor, it would appear that water molecules are introduced into $LiBH_4$ lattice as structural water. The absorbed volume measured by the Sieverts apparatus was almost identical to that calculated by sample mass change before and after measurements were taken. This implies that the side reactions which result in hydrogen emission can be ignored.

The crystal structure of hydrated $LiBH_4$ was analyzed by XRD (Fig. 2). Only a monoclinic (space group $P2_1/c$) pattern attributed to $LiBH_4 \cdot H_2O$ was obtained with the following lattice parameters: $a = 6.263$ Å, $b = 6.302$ Å, $c = 10.004$ Å and $\beta = 117.61^\circ$. The XRD pattern indicates that $LiBH_4 \cdot H_2O$ was produced without any decomposition [19].

3.2. Thermal stability of hydrated $LiBH_4$

As expected, the hydration–dehydration reaction of $LiBH_4$ was sensitive to temperature. Fig. 3 shows the DSC curves of hydrated $LiBH_4$. As shown in Fig. 3(a), the sample was first heated from −20 °C to 150 °C, then cooled down to −20 °C. This cycle was repeated 2 times. To take this measurement, a hermetic pan with a pin hole was used to avoid any increase in pressure due to dehydration. An endothermic peak was observed at around 50 °C in the 1st heating process. This peak can be attributed to the dehydration reaction of hydrated $LiBH_4$. The dehydration temperature and its enthalpy were estimated to be 53 °C and 21 kJ · mol–H_2O^{-1}, respectively. The temperature was further increased, and a large exothermic peak appeared above 60 °C. At this temperature, $LiBH_4$ presumably reacted with dehydrated water vapor and decomposed. According to J. P. Goudon et al., it is likely that either lithium metaborates or its hydrates were produced [16].

A DSC curve for a lower-temperature cycling using a pan without a pin hole is presented in Fig. 3(b). The sample was heated from −20 °C

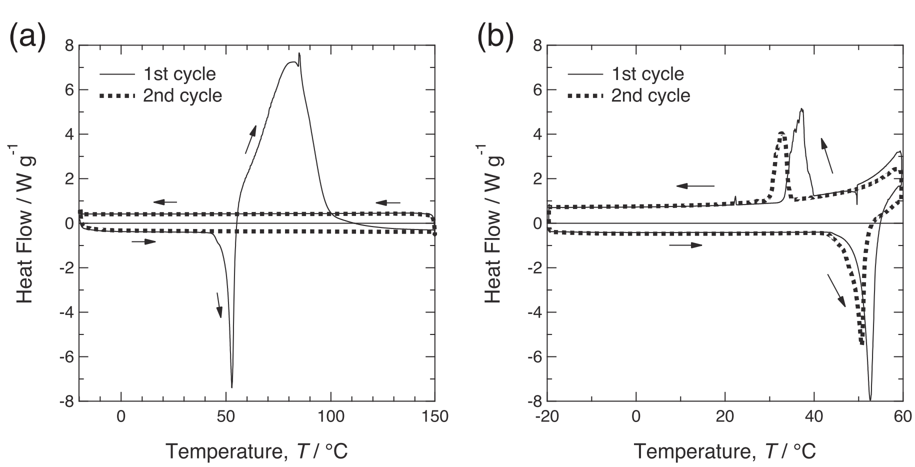

Fig. 3. Temperature dependences of heat flow measured by DSC. (a) A higher-temperature cycling measurement. The sample was heated from −120 °C to 150 °C, then cooled to −20 °C, which was repeated 2 times. (b) A lower-temperature cycling measurement, which was the same as (a), except in the maximum temperature of 60 °C.

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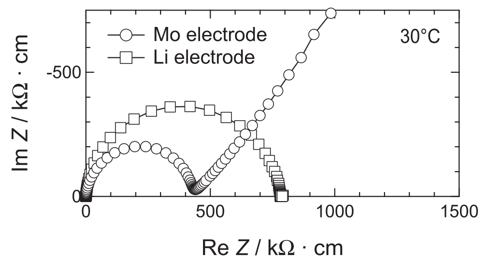

Fig. 4. Nyquist plots of Mo | hydrated $LiBH_4$ | Mo symmetrical cell and Li | hydrated $LiBH_4$ | Li symmetrical cell.

to 60 °C, then cooled down to −20 °C. The cycle was repeated 2 times. As with the higher-temperature cycling (Fig. 3(a)), an endothermic peak was observed at around 50 °C during the 1st heating process. In the 1st cooling process, an exothermic peak was observed at around 35 °C, while none was observed in the 1st higher-temperature cooling cycling. The exothermic peak and the endothermic peak were also observed in the 2nd cycle. This suggests that $LiBH_4$ can be reversibly hydrated and dehydrated below 60 °C.

3.3. Lithium-ion conduction in hydrated $LiBH_4$

The electrical conductivity of hydrated $LiBH_4$ was evaluated by AC impedance measurements. To investigate conductive species, two types of symmetrical cells were used, Mo | hydrated $LiBH_4$ | Mo and Li | hydrated $LiBH_4$ | Li. Nyquist plots of these cells at 30 °C are shown in Fig. 4. When using molybdenum electrodes, which are irreversible against lithium ions, a semicircle followed by a Warburg-type impedance was observed. Using an equivalent circuit model, the capacitance was estimated at 10^{-10} F, suggesting that the semicircle can be attributed to bulk conduction resistance. The same responses were observed at the other measured temperatures.

When a lithium electrode was used, only a semicircle was observed with a capacitance value estimated in the order of 10^{-10} F, indicating bulk resistance. The absence of Warburg impedance suggests that the lithium metal functions well as a reversible electrode and that there is no interfacial resistance between $LiBH_4$ and the lithium metal [15]. In fact, the two-probe DC conductivity measurement of the cell with lithium electrodes was almost identical to that measured by the AC

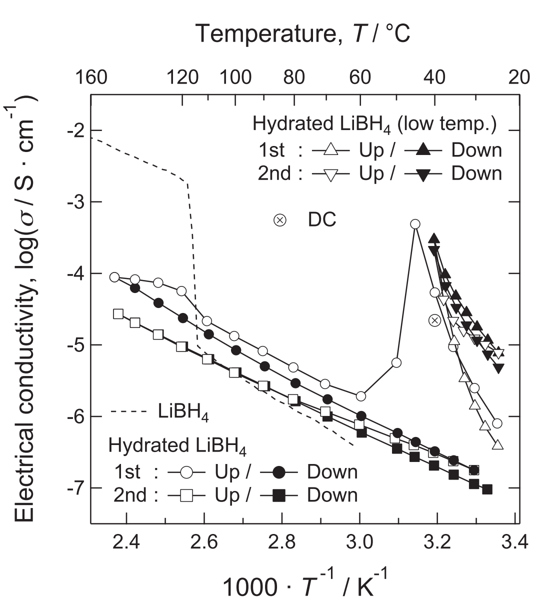

Fig. 5. Arrhenius plot of electrical conductivity of hydrated $LiBH_4$. $LiBH_4$ data was cited from Matsuo et al. [12]. The DC conductivity was also plotted.

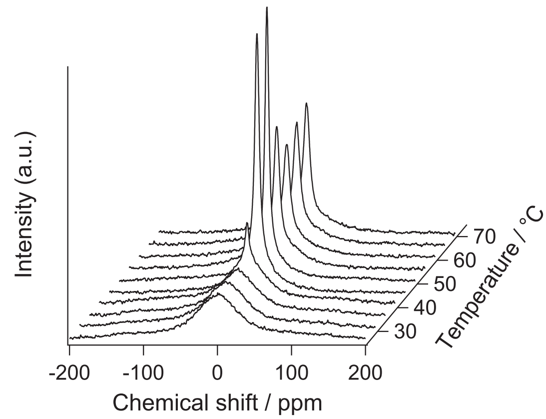

Fig. 6. ^{7}Li NMR spectra of hydrated $LiBH_4$ from 25 to 70 °C.

impedance technique. It should be noted, however, that the semicircle created by the cell using the lithium electrode was slightly larger than that created by the molybdenum electrode. In addition, the increase in the diameter of the semicircle over time indicates that side reactions between lithium metal electrodes and hydrated $LiBH_4$ take place.

The temperature dependence of electrical conductivity was investigated by AC impedance measurement using the molybdenum electrode. The Arrhenius plot of the electrical conductivity calculated from the bulk resistance is shown in Fig. 5. The DC conductivity measured by using lithium electrodes is also plotted in Fig. 5. A maximum value of 4.89×10^{-4} S · cm^{-1} was attained at 45 °C. This conductivity is comparable with that of the high temperature phase of $LiBH_4$. The activation energy of 253 kJ · mol^{-1} obtained between 25 °C and 45 °C was much higher than that of the low temperature and high temperature phase of $LiBH_4$ (66.6 kJ · mol^{-1} and 51.1 kJ · mol^{-1}, respectively [12]). This suggests that the mechanism of lithium-ion conduction in hydrated $LiBH_4$ differs from that of $LiBH_4$. The dramatic decrease in conductivity above 50 °C is coincident with the point of dehydration, as observed by DSC (Fig. 3). This implies that structural water molecules play an important role in lithium-ion conduction in hydrated $LiBH_4$. In addition to DSC measurements, AC impedance measurements were carried out below 40 °C to confirm the reversibility in the change in conductivity. The Arrhenius plot taken below 40 °C is displayed in Fig. 5 by triangle markers. It was confirmed that the enhancement of electrical conductivity showed reversibility when temperature changed below the water desorption temperature. Meanwhile, hydrated $LiBH_4$ showed no phase transition above the dehydration temperature. This is presumably due to the distortion of the atomic arrangement together with hydration and dehydration. In fact, the XRD patterns and the broadening of the peaks in the Raman spectra indicate that the $LiBH_4$ became distorted after the AC impedance measurement at elevated temperatures (not shown). Recently, Blanchard et al. have reported that nanoconfinement affects the phase transition temperature of $LiBH_4$ [20]. This suggests that the phase transition of $LiBH_4$ is sensitive to its local structure at the nanoscale level.

To confirm lithium-ion conduction, ^{7}Li nuclear magnetic resonance spectroscopy was carried out. Fig. 6 shows the ^{7}Li NMR spectra of hydrated $LiBH_4$ from 25 to 70 °C. The motional narrowing indicates that lithium-ion conduction was remarkable at around 45 °C. The broader peaks also observed shows that slow lithium ions are likely to exist with fast ions. The reciprocal values of the full width at the half maximum (FWHM) of the ^{7}Li NMR spectra were plotted as a function of temperature in Fig. 7. This response was consistent with the temperature dependence of electrical conductivity, as shown in Fig. 5. Therefore, it can be concluded that hydration enhances the lithium-ion conductivity of $LiBH_4$.

We propose that the motion of lithium ions in hydrated $LiBH_4$ depends on their interaction with the vibration of water molecules.

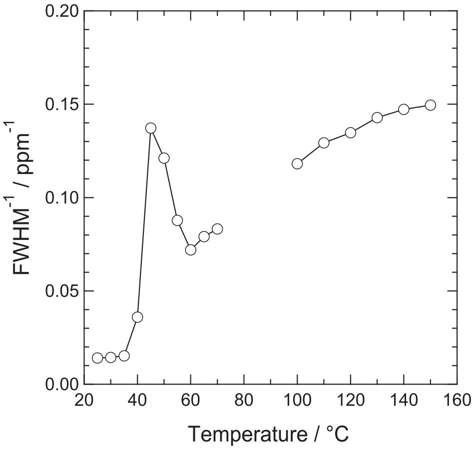

Fig. 7. Reciprocal values of FWHM of ^{7}Li NMR spectra as a function of temperature.

Lithium-ion conduction in $LiBH_4$ is deeply associated with vibrations and the rotation of Li^+ and BH_4^- [21,22]. However, a different mechanism of lithium-ion conduction appears in hydrated $LiBH_4$ based on a much higher activation energy. As mentioned before, the enhancement of lithium-ion conductivity was remarkable near dehydration temperature. In addition, from the viewpoint of crystal structure, Li–H_2O interaction appears to be strong in $LiBH_4 \cdot H_2O$. In the $LiBH_4 \cdot H_2O$ lattice, Li^+ is surrounded by two BH_4^- and two H_2O molecules. The shorter distance between Li and O in H_2O (2.04 Å in average) than between Li and B in BH_4^- (4.88 Å in average) suggests that the H_2O molecule strongly interacts with Li. In other words, the enhancement of lithium-ion conductivity may be mediated by the motion of structural water.

4. Conclusions

$LiBH_4 \cdot H_2O$ was formed by water vapor introduction without decomposition. The structural water can be dehydrated at 53 °C with the enthalpy change of 21 kJ · mol–H_2O^{-1} and rehydrates at around 35 °C reversibly.

Hydrated $LiBH_4$ showed a much higher electrical conductivity around R.T. (4.89×10^{-4} S · cm^{-1} at 45 °C) than $LiBH_4$. AC impedance measurements using a lithium electrode and the motional narrowing of the ^{7}Li NMR indicate that the major carrier was the lithium-ion. These results strongly suggest that the dynamics of structural water plays an important role in lithium-ion conductivity.

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